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Assignment Kubernetes Basics

Question 1

```
nidhi@nidhi-Predator-PHN16-72:~$ minikube start
W0908 21:15:31.674167 16008 main.go:294] Unable to resolve the current Docker CLI context "default": context "default": context not found: open /home/nidhi/.docker/contexts/meta/37a8
ee1ce19687d132fe29051dca629d164e2c4958ba141d5f4133a33f0688f/meta.json: no such file or directory
W0908 21:15:31.674208 16008 main.go:295] Try running 'docker context use default' to resolve the above error
🐳 minikube v1.39.0 on Ubuntu 24.04
🔧 Using the docker driver based on existing profile
🔥 Starting "minikube" primary control-plane node in "minikube" cluster
📶 Pulling base image v0.0.51 ...
🔄 Restarting existing docker container for "minikube" ...
🔧 Preparing Kubernetes v1.37.0 on containerd 2.3.4 ...
🔧 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: storage-provisioner, default-storageclass
🏠 Done! kubectrl is now configured to use "minikube" cluster and "default" namespace by default
nidhi@nidhi-Predator-PHN16-72:~$ kubectl get nodes
NAME                STATUS   ROLES    AGE   VERSION
minikube            Ready    control-plane 2d10h v1.37.0
nidhi@nidhi-Predator-PHN16-72:~$ kubectl version --client
Client Version: v1.36.3
Kustomize Version: v5.8.1
nidhi@nidhi-Predator-PHN16-72:~$
```

Question 2

```
nidhi@nidhi-Predator-PHN16-72:~$ kubectl run myweb --image=nginx
pod/myweb created
nidhi@nidhi-Predator-PHN16-72:~$ kubectl get pods
NAME    READY   STATUS    RESTARTS   AGE
myweb   1/1     Running   0          7s
nidhi@nidhi-Predator-PHN16-72:~$ kubectl describe pod myweb
Name:          myweb
Namespace:     default
Priority:       0
Service Account: default
Node:          minikube/192.168.49.2
Start Time:    Tue, 08 Sep 2026 21:24:07 +0530
Labels:        run=myweb
Annotations:   <none>
Status:        Running
IP:            10.244.0.6
IPs:
  IP: 10.244.0.6
Containers:
  myweb:
    Container ID:  containerd://14f641b6dca58633cc5084e6ea779a2b3e286ab6b8a289314ca0d53ac91731d8
    Image:         nginx
    Image ID:      docker.io/library/nginx@sha256:05b8cb60c354a44ab824ea6e7dc69b46d50762cdbe728a347a5b656e6fb3d7c4
    Port:         <none>
    Host Port:     <none>
    State:         Running
      Started:     Tue, 08 Sep 2026 21:24:10 +0530
    Ready:         True
    Restart Count: 0
    Environment:   <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-x9spm (ro)
Conditions:
  Type                               Status
  PodReadyToStartContainers         True
  Initialized                        True
  Ready                             True
  ContainersReady                   True
  PodScheduled                      True
Volumes:
  kube-api-access-x9spm:
    Type:          Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName:  kube-root-ca.crt
    Optional:       false
    DownwardAPI:    true
QoS Class:         BestEffort
Node-Selectors:    <none>
Tolerations:       node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
```

```

Normal Pulling 11s kubelet spec.containers{myweb}: Pulling image "nginx"
Normal Pulled 9s kubelet spec.containers{myweb}: Successfully pulled image "nginx" in 2.218s (2.218s including waiting). Image size: 66343926 bytes.
Normal Created 9s kubelet spec.containers{myweb}: Container created
Normal Started 9s kubelet spec.containers{myweb}: Container started
nidhi@nidhi-Predator-PHN16-72:~$ kubectl logs myweb
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/09/08 15:54:10 [notice] 1#1: using the "epoll" event method
2026/09/08 15:54:10 [notice] 1#1: nginx/1.31.5
2026/09/08 15:54:10 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/09/08 15:54:10 [notice] 1#1: OS: Linux 7.0.0-28-generic
2026/09/08 15:54:10 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/09/08 15:54:10 [notice] 1#1: start worker processes
2026/09/08 15:54:10 [notice] 1#1: start worker process 29
2026/09/08 15:54:10 [notice] 1#1: start worker process 30
2026/09/08 15:54:10 [notice] 1#1: start worker process 31
2026/09/08 15:54:10 [notice] 1#1: start worker process 32
2026/09/08 15:54:10 [notice] 1#1: start worker process 33
2026/09/08 15:54:10 [notice] 1#1: start worker process 34
2026/09/08 15:54:10 [notice] 1#1: start worker process 35
2026/09/08 15:54:10 [notice] 1#1: start worker process 36
2026/09/08 15:54:10 [notice] 1#1: start worker process 37
2026/09/08 15:54:10 [notice] 1#1: start worker process 38
2026/09/08 15:54:10 [notice] 1#1: start worker process 39
2026/09/08 15:54:10 [notice] 1#1: start worker process 40
2026/09/08 15:54:10 [notice] 1#1: start worker process 41
2026/09/08 15:54:10 [notice] 1#1: start worker process 42
2026/09/08 15:54:10 [notice] 1#1: start worker process 43
2026/09/08 15:54:10 [notice] 1#1: start worker process 44
2026/09/08 15:54:10 [notice] 1#1: start worker process 45
2026/09/08 15:54:10 [notice] 1#1: start worker process 46
2026/09/08 15:54:10 [notice] 1#1: start worker process 47
2026/09/08 15:54:10 [notice] 1#1: start worker process 48
2026/09/08 15:54:10 [notice] 1#1: start worker process 49
2026/09/08 15:54:10 [notice] 1#1: start worker process 50
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2026/09/08 15:54:10 [notice] 1#1: start worker process 53
2026/09/08 15:54:10 [notice] 1#1: start worker process 54
2026/09/08 15:54:10 [notice] 1#1: start worker process 55
2026/09/08 15:54:10 [notice] 1#1: start worker process 56
nidhi@nidhi-Predator-PHN16-72:~$

```

Question 3

```

nidhi@nidhi-Predator-PHN16-72:~$ kubectl delete pod myweb
pod "myweb" deleted from default namespace
nidhi@nidhi-Predator-PHN16-72:~$ kubectl get pods
No resources found in default namespace.
nidhi@nidhi-Predator-PHN16-72:~$

```

Question 4

```
nidhi@nidhi-Predator-PHN16-72:~$ kubectl create deployment web --image=nginx --replicas=2
deployment.apps/web created
nidhi@nidhi-Predator-PHN16-72:~$ kubectl get deployments
NAME    READY   UP-TO-DATE   AVAILABLE   AGE
web     2/2     2            2           16s
nidhi@nidhi-Predator-PHN16-72:~$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
web-5fc9f4bf66-5htgb               1/1     Running   0           24s
web-5fc9f4bf66-8l8bk               1/1     Running   0           24s
nidhi@nidhi-Predator-PHN16-72:~$ kubectl delete pod web-5fc9f4bf66-5htgb
pod "web-5fc9f4bf66-5htgb" deleted from default namespace
nidhi@nidhi-Predator-PHN16-72:~$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
web-5fc9f4bf66-8l8bk               1/1     Running   0           76s
web-5fc9f4bf66-crkl               1/1     Running   0           7s
nidhi@nidhi-Predator-PHN16-72:~$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
web-5fc9f4bf66-8l8bk               1/1     Running   0           2m25s
web-5fc9f4bf66-crkl               1/1     Running   0           76s
nidhi@nidhi-Predator-PHN16-72:~$
```

```
nidhi@nidhi-Predator-PHN16-72:~$ kubectl scale deployment web --replicas=4
deployment.apps/web scaled
nidhi@nidhi-Predator-PHN16-72:~$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
web-5fc9f4bf66-8l8bk               1/1     Running   0           4m7s
web-5fc9f4bf66-9rlk4               1/1     Running   0           11s
web-5fc9f4bf66-crkl               1/1     Running   0           2m58s
web-5fc9f4bf66-r85wp               1/1     Running   0           11s
nidhi@nidhi-Predator-PHN16-72:~$ kubectl get deployments
NAME    READY   UP-TO-DATE   AVAILABLE   AGE
web     4/4     4            4           4m23s
nidhi@nidhi-Predator-PHN16-72:~$
```

Question 5

```
nidhi@nidhi-Predator-PHN16-72:~$ kubectl expose deployment web --port=80 --type=NodePort
Error from server (AlreadyExists): services "web" already exists
nidhi@nidhi-Predator-PHN16-72:~$ kubectl get services
NAME                TYPE        CLUSTER-IP      EXTERNAL-IP  PORT(S)          AGE
kubernetes           ClusterIP   10.96.0.1        <none>       443/TCP          2d11h
my-db-postgresql     ClusterIP   10.107.148.239   <none>       5432/TCP         2d8h
my-db-postgresql-hl ClusterIP   None             <none>       5432/TCP         2d8h
test-app             NodePort    10.108.133.1     <none>       80:30973/TCP     2d9h
web                  NodePort    10.98.41.87      <none>       80:31810/TCP     2d9h
nidhi@nidhi-Predator-PHN16-72:~$ minikube service web
W0908 21:34:38.268803 32565 main.go:294] Unable to resolve the current Docker CLI context "default": context "default": context not found: open /home/nidhi/.docker/contexts/n
eec1ce19687d132fe29051dca629d164e2c4958ba141d5f4133a33f0688f/meta.json: no such file or directory
W0908 21:34:38.268825 32565 main.go:295] Try running `docker context use default` to resolve the above error
```

NAMESPACE	NAME	TARGET PORT	URL
default	web	80	http://192.168.49.2:31810

 Opening service default/web in default browser...

```
nidhi@nidhi-Predator-PHN16-72:~$ Opening in existing browser session.
curl -i http://192.168.49.2:31810
HTTP/1.1 200 OK
Server: nginx/1.31.5
Date: Tue, 08 Sep 2026 16:05:34 GMT
Content-Type: text/html
Content-Length: 896
Last-Modified: Wed, 02 Sep 2026 11:17:13 GMT
Connection: keep-alive
ETag: "6a9805b9-380"
Accept-Ranges: bytes

<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
nidhi@nidhi-Predator-PHN16-72:~$
```